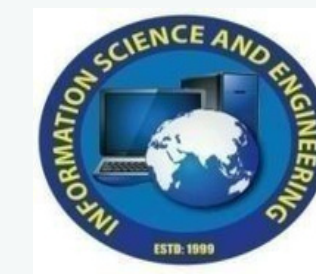




THE NATIONAL INSTITUTE OF ENGINEERING



Department Of Information Science And Engineering

MINOR PROJECT- BIS586

Group C9

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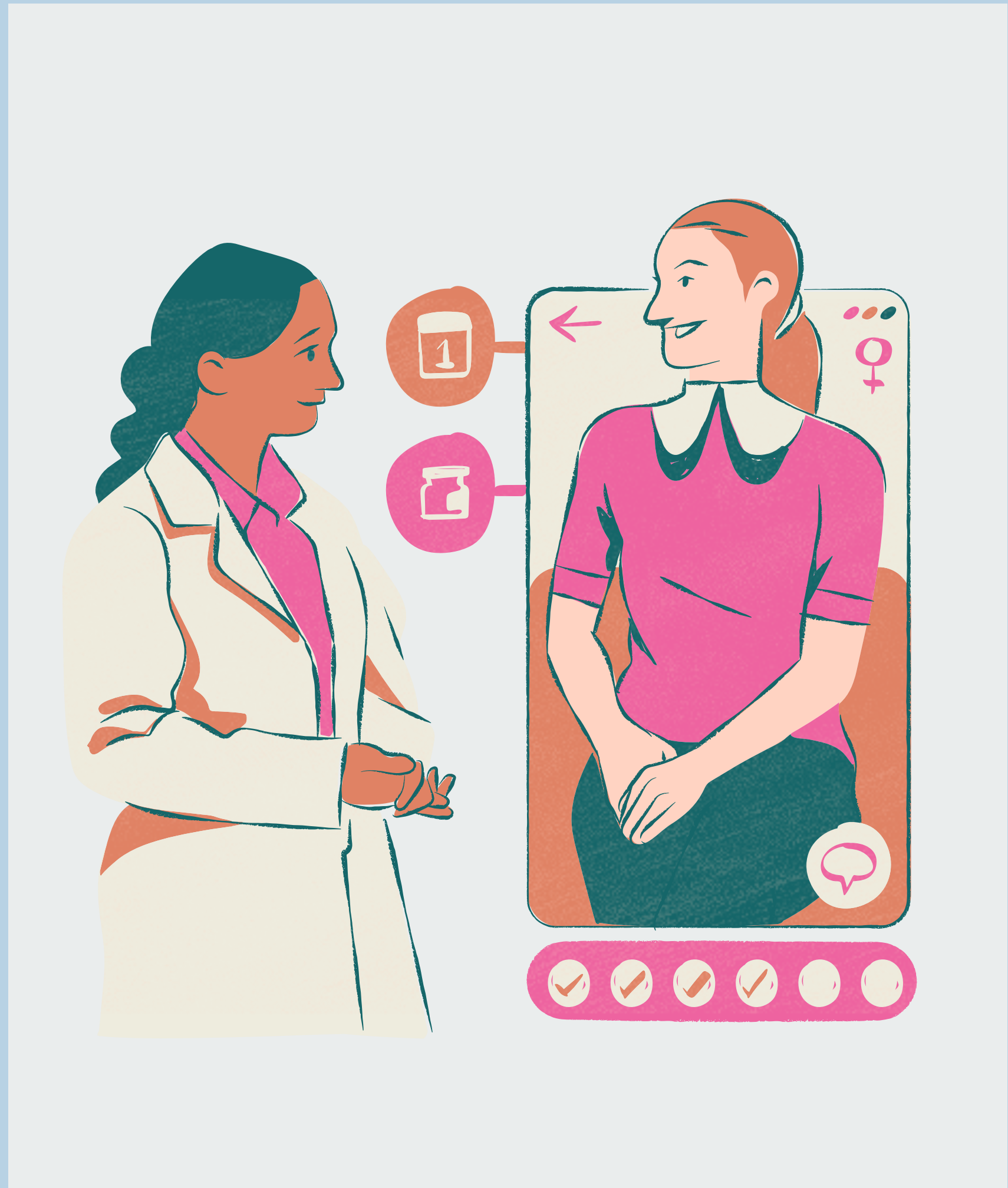
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An Integrated Platform for medical consultation and digital health records management



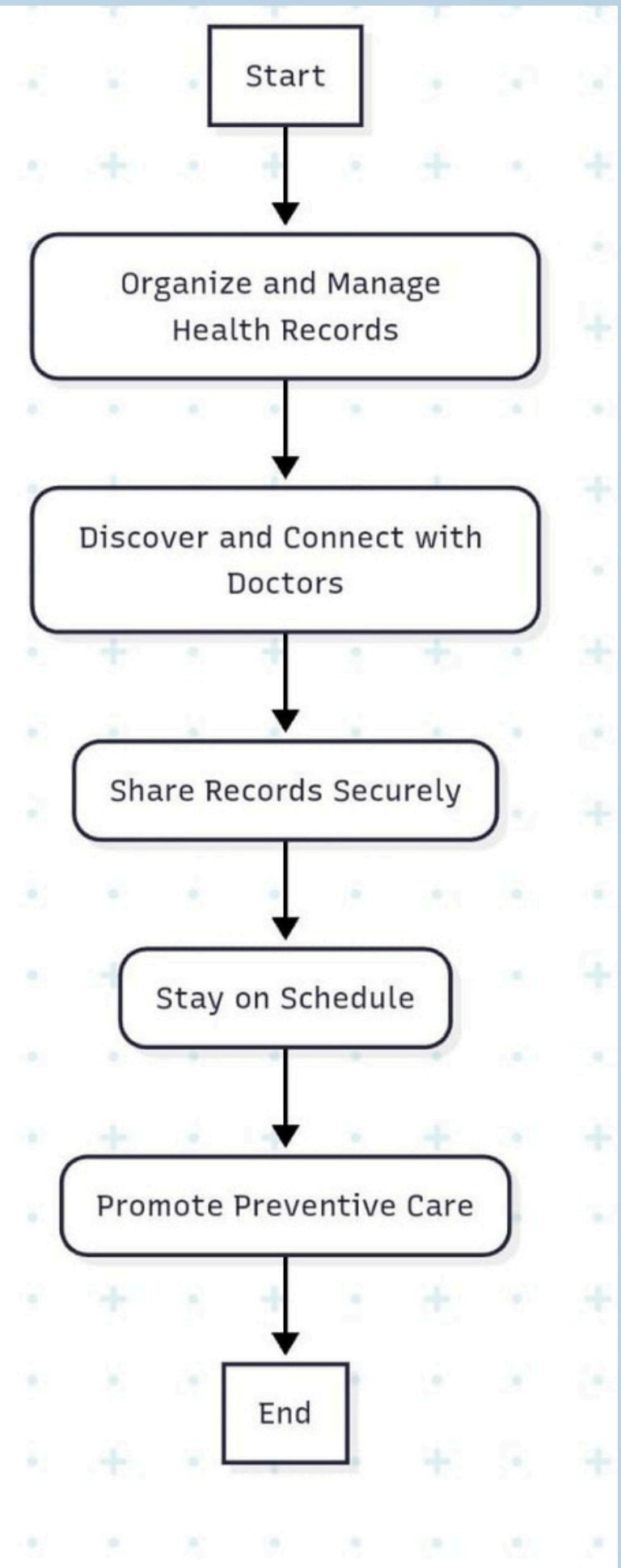
Abstract

This project introduces an innovative, patient-centric digital platform designed to restore control over personal health information. At its core, the application provides a highly secure, encrypted vault for medical records, architected with a zero-knowledge protocol to guarantee absolute user privacy. Complementing this, the platform streamlines the entire consultation journey, from discovering trusted medical professionals to scheduling appointments and securely sharing information. By integrating these critical functions, we aim to deliver a seamless, empowering, and confidential healthcare management experience.



Introduction

Our project, "An Integrated Platform for Medical Consultation and Digital Health Records Management," tackles the problem of scattered and insecure health information. We're building a secure, zero-knowledge platform where users fully control encrypted medical records, ensuring privacy even from administrators. Along with record management, the app enables hospital and doctor discovery, appointment booking, and offers curated preventive health tips.



Objectives

The primary goal of this project is to build a secure, all-in-one platform for personal health management. Our key objectives are to allow patients to:

- **Organize and Manage Health Records:** Create personal accounts to securely store, organize, and manage their own digital health records.
- **Discover and Connect with Doctors:** Search for hospitals and doctors across various cities, view detailed information on their qualifications and specialties, and book appointments.
- **Share Records Securely:** Send health records directly to doctors using temporary email links for consultations.
- **Stay on Schedule:** Receive timely reminders for upcoming appointments.
- **Promote Preventive Care:** Stay informed with regularly updated health facts and tips displayed on the platform's homepage.

Broad Area of Project Work

Broad area of our project is Full-stack development

Type of Project: Application-based Minor Project

Frontend:

React.js – dynamic UI for patients & doctors

Backend:

Spring Boot 3 – APIs, logic & secure processing



Database:

MongoDB – scalable, schema-flexible storage

File Storage:

Cloudinary – encrypted record storage

Deployment:

Render – cloud-based hosting & scaling

Literature Survey

Problems:

- Delay in accessing medical reports.
- No standardized doctor appointments in emergencies.
- Lack of timely medication reminders.

01. Fully encrypted vault for medical reports.

02. Browse hospitals, view doctors with details, and choose the right specialist.
Book doctors directly and share reports instantly via email.

03. Eliminates report-hunting and speeds up consultations.

Literature Review

No single platform offers appointments + doctor/hospital info + EHR/report management + secure sharing.

Our platform provides

- Appointment scheduling
- Hospital & doctor discovery
- Health records storage & sharing (Electronic Health Record integration)
- Easy report transfer via email

Reference papers used :

- Link : [Health Care Management: Principles and Practice](#)
- Link : [Personal Health Records: A Systematic Literature Review.](#)
- Link : [Effects of Electronic Health Record Implementation and Barriers to EHR Adoption: A Scoping Review.](#)
- Link : [Integrated Personal Health Record \(PHR\) security: Requirements and mechanisms.](#)





Problem Formation



01.

Data Fragmentation & Inaccessibility:

Patients' critical health histories are scattered across different clinics and hospitals, often in physical formats, making it nearly impossible to access a complete, unified record during a medical consultation.

02.

Lack of Data Sovereignty & Security Risks:

Conventional digital platforms store sensitive health information in centralized databases, leaving patient data vulnerable to breaches and accessible to platform administrators, thereby stripping patients of true ownership.

03.

Inefficient Consultation Lifecycle:

The journey from finding a qualified doctor in an unfamiliar city to booking an appointment and ensuring the doctor has the correct prior medical context is disjointed, time-consuming, and prone to communication gaps.

Scope



In Scope:

Development of a patient-facing web interface using React.js.

Building a robust backend API using Spring Boot 3.

Integration with a MongoDB database for application data and an S3-compatible service for encrypted file storage.

The application will be deployed and fully functional on the Render platform.

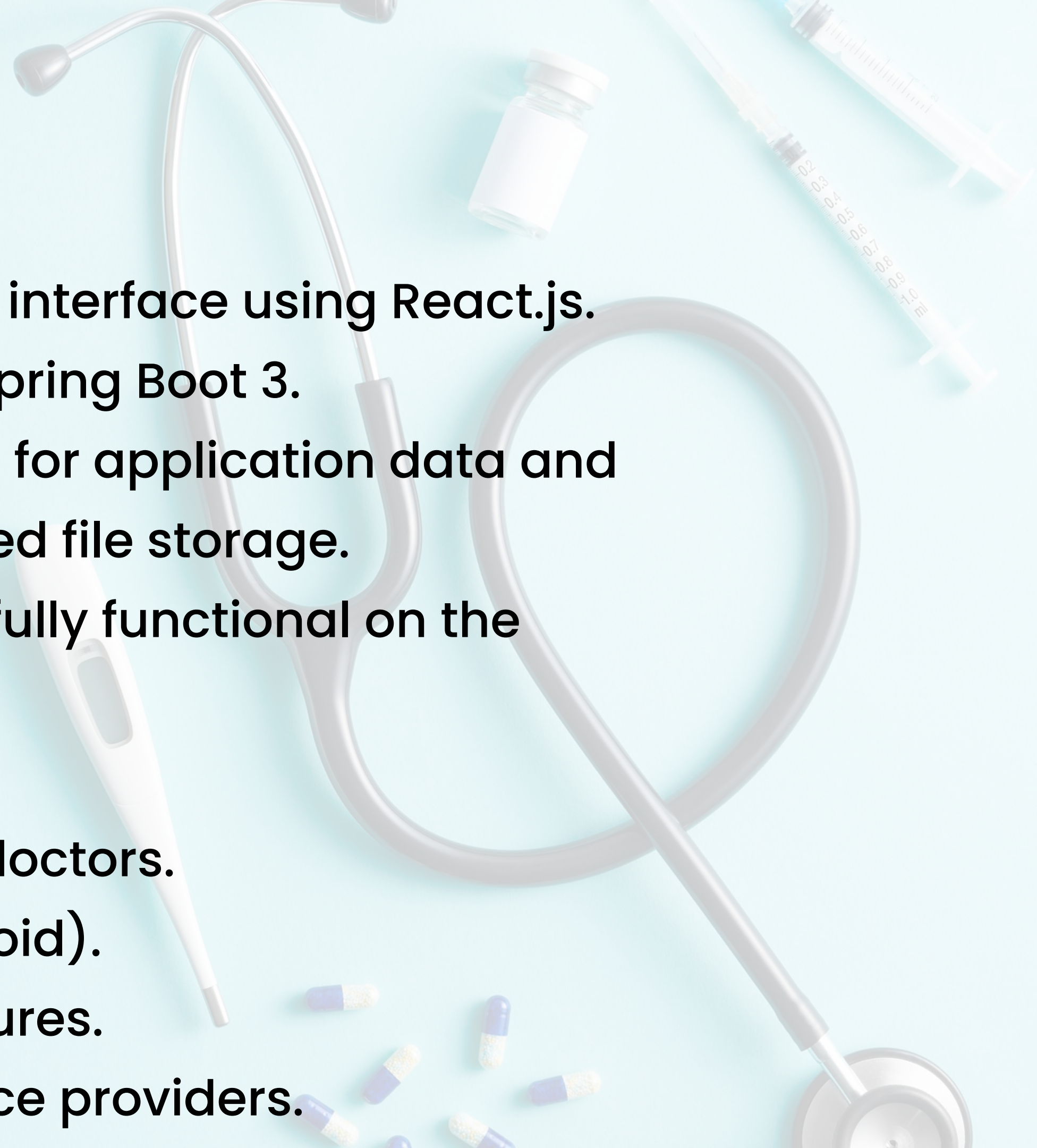
Out of Scope for this Phase:

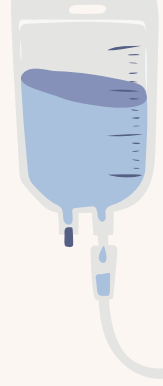
A dedicated portal or application for doctors.

Native mobile applications (iOS/Android).

Real-time video teleconsultation features.

Integration with pharmacy or insurance providers.





Design

Architecture:

React.js frontend for users → Spring Boot backend for logic → MongoDB for data + clouinary for encrypted files + mailersend for email.

Core Modules:

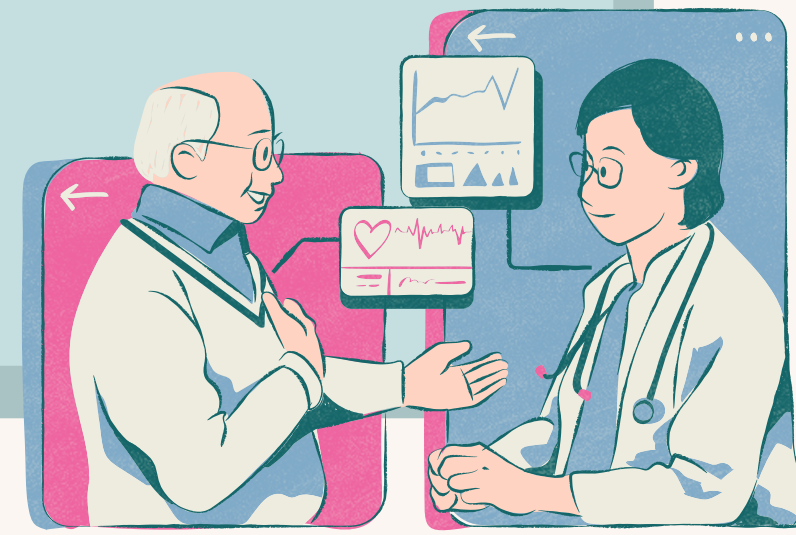
User (profiles & records), Appointments (booking & reminders), Health Records (upload & retrieve), Search (doctors/hospitals)

Database Layer:

Stores patients, doctors, health records, appointments, and system logs for tracking activities

Security Layer:

Zero-knowledge encryption ensures only users and the doctors that they send their records to can view records



Thank you



Summary of Your Unique Value Proposition

When you put it all together, the uniqueness of your project becomes clear:

While competitors like Practo and Apollo 24/7 offer the convenience of digital records, they operate on a model of platform-controlled data. Our project introduces a paradigm shift by delivering verifiable patient sovereignty. We achieve this through a unique combination of foundational privacy with our zero-knowledge architecture and functional control with our secure, temporary email sharing links. We are not just giving patients a place to store their records; we are giving them the exclusive keys to their own vault and complete control over who gets a temporary look inside.